

Polynomial Regression Assignment Report

- **Name:** Pragun K Kirani
- **Roll Number:** BT2024176
- **Github** <https://github.com/pragunkk/MLregressionAssignment1>

Introduction

In this assignment, I built polynomial regression models for two distinct phases of a multi-stage geothermal power plant expansion project. Each phase presented unique challenges and required me to build a separate model to predict the target variable **y** given a set of features. My primary objective was to obtain accurate predictions on the provided test datasets by figuring out the optimal polynomial degree and selecting the appropriate features.

I used the provided datasets ([BT2024176_train_var1.csv](#) and [BT2024176_train_var2.csv](#)) to train my models, and I generated predictions on [BT2024176_test_var1.csv](#) and [BT2024176_test_var2.csv](#).

Approach and Methodology

To select the optimal polynomial degree, I took a rigorous empirical approach. I performed **5-Fold Cross-Validation** over a large range of polynomial degrees. My goal was to maximize predictive performance (R^2 Score) while minimizing Mean Squared Error (MSE), making sure to avoid both underfitting and overfitting.

Phase 1: Power Plant Steam Turbine Optimization (var1)

Problem Description

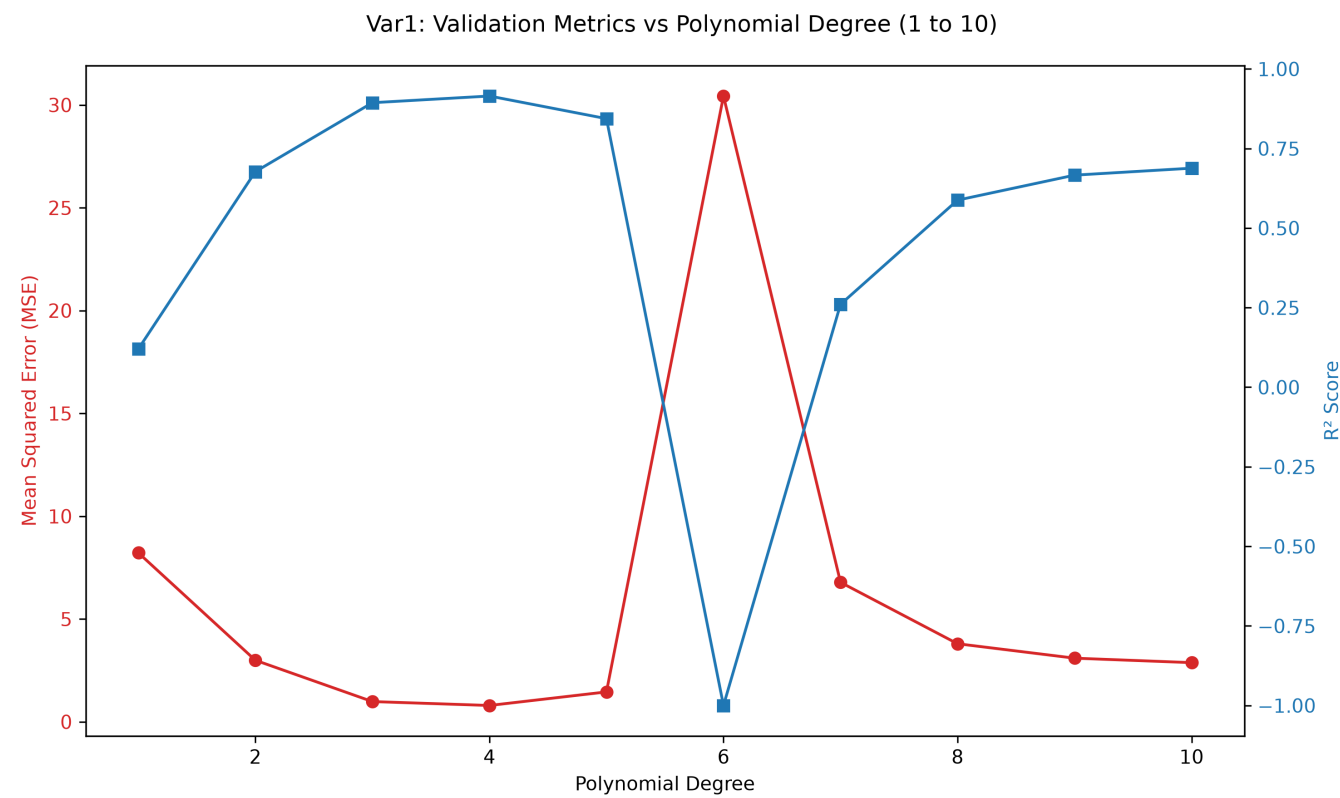
The first phase involves predicting the Net Power Score (**y**) based on 6 operational parameters (**x1** to **x6**) representing percentage deviations in the power plant settings.

Hyperparameter Tuning Results (Degrees 1 to 10)

Using all 6 features, I evaluated models from degree 1 through 10 using 5-Fold Cross-Validation.

Degree	MSE	R ² Score
1	8.2186	0.1202
2	2.9925	0.6773
3	0.9841	0.8939
4	0.7940	0.9149
5	1.4540	0.8438
6	123.0929	-12.8296
7	6.7832	0.2604

Degree	MSE	R ² Score
8	3.7966	0.5878
9	3.0945	0.6663
10	2.8772	0.6882



Model Selection and Rationale

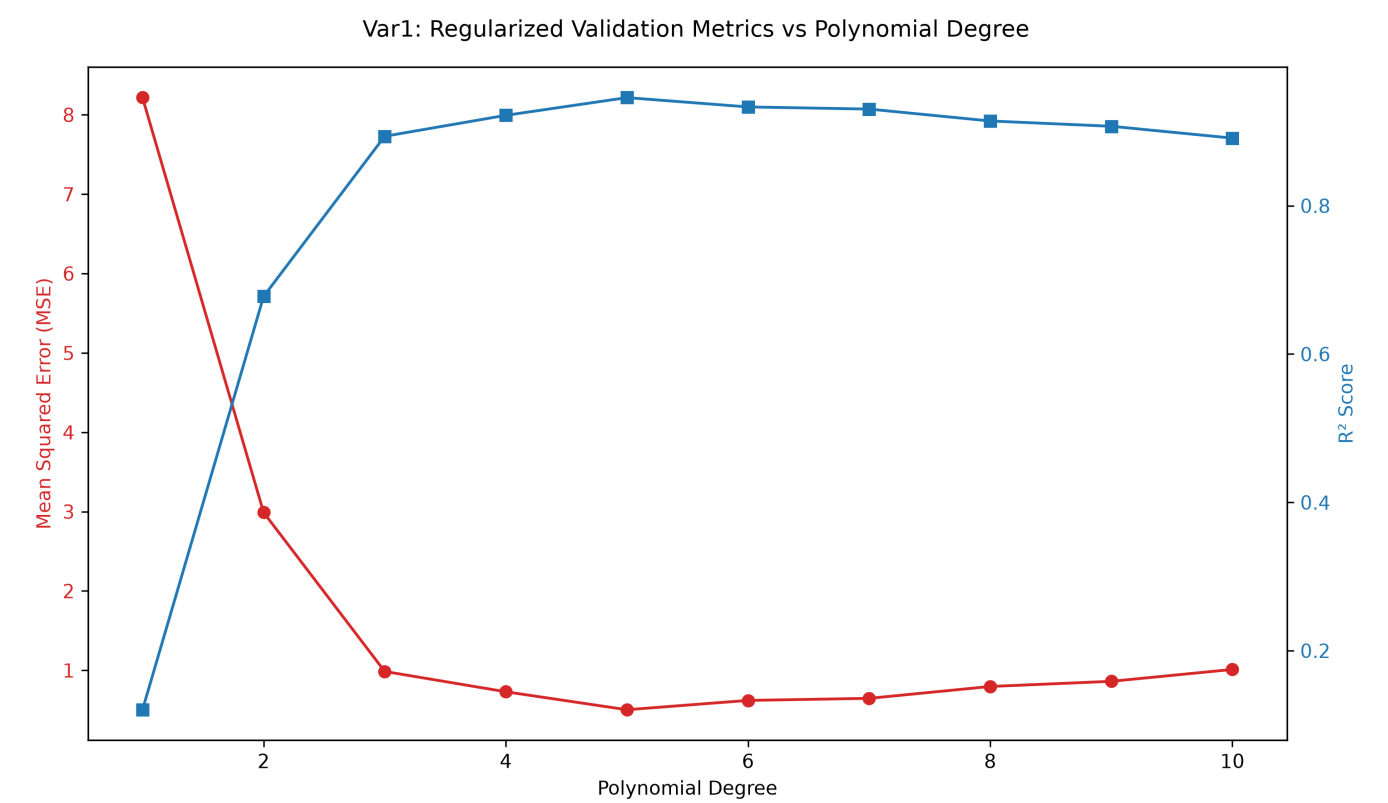
Based on the table and the graph, I observed the following:

- 1. **Underfitting at Low Degrees:** At degrees 1 and 2, my model underfit the data ($R^2 = 0.1202$ and 0.6773).
- 2. **Optimal Fit at Degree 4:** As the degree increased, MSE decreased and R^2 improved, peaking at **Degree 4** ($MSE = 0.7940$, $R^2 = 0.9149$).
- 3. **Overfitting at High Degrees:** Beyond degree 4, the validation MSE sharply increased, and the R^2 score dropped drastically, entering severe negative values at degree 6. This was a classic indication of the model overfitting to noise in the training set and failing to generalize. Although the metrics stabilized somewhat after degree 6, they never approached the performance of degree 4.

Regularization Testing (Ridge)

To further improve performance and prevent overfitting at higher degrees, I also tested Ridge Regularization combined with Standard Scaling across all degrees.

- **OLS Best:** Degree 4 (Val $R^2 = 0.9149$)
- **Ridge Best:** Applying Ridge regression ($\alpha \approx 11.2884$) allowed the model to safely utilize **Degree 5** polynomial features without overfitting, improving the peak validation R^2 score to **0.9458**.



Conclusion for Phase 1: I decided the optimal model is a **Ridge Regularized Polynomial model of Degree 5 using all 6 features**, as it yields superior generalization.

Phase 2: Subterranean Thermal Reservoir Mapping (var2)

Problem Description

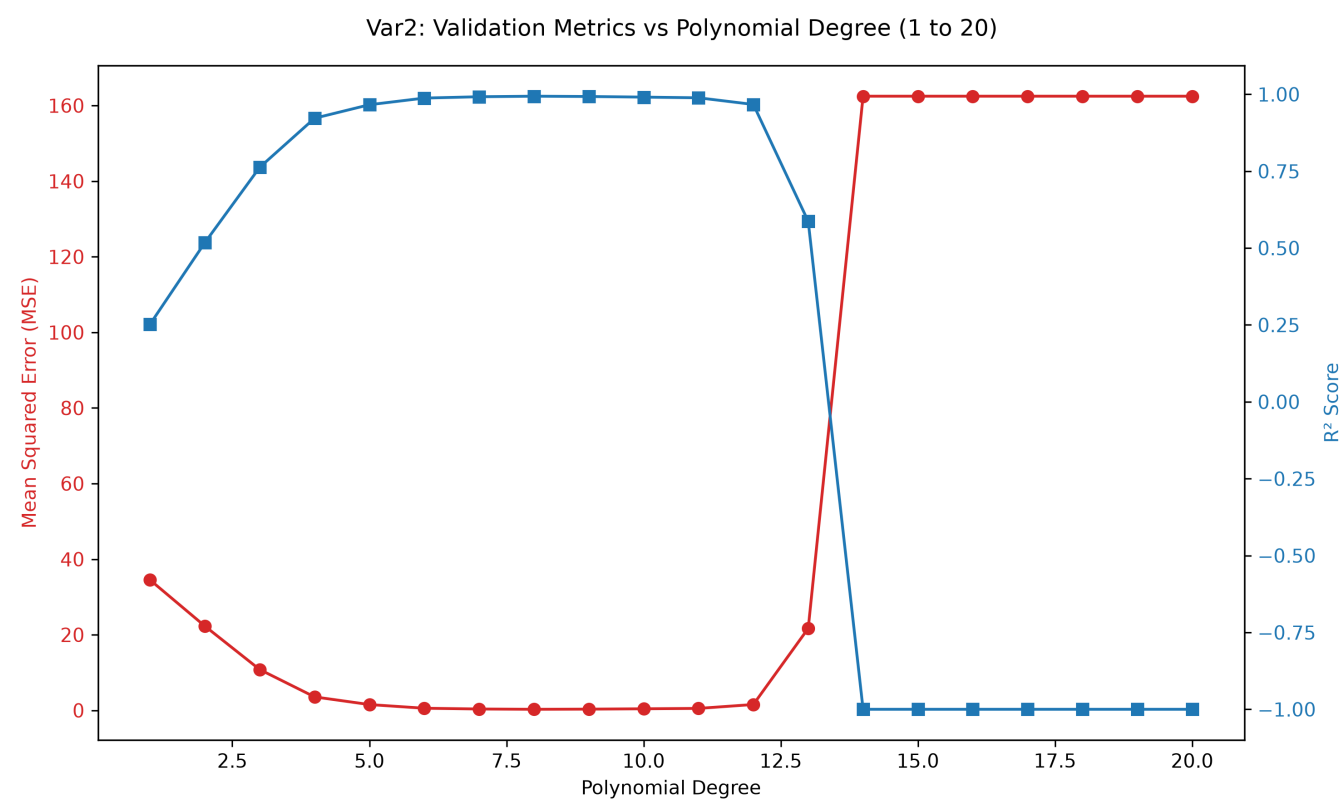
The second phase requires predicting a Thermal Anomaly Score (**y**) based on 3 spatial coordinate offsets (**x1**, **x2**, **x3**).

Hyperparameter Tuning Results (Degrees 1 to 20)

Using all 3 spatial features, I evaluated models from degree 1 through 20 using 5-Fold Cross-Validation.

Degree	MSE	R² Score
1	34.5026	0.2522
2	22.2891	0.5177
3	10.7983	0.7626
4	3.5661	0.9221
5	1.5367	0.9660
6	0.5637	0.9876
7	0.3623	0.9920
8	0.2854	0.9938

Degree	MSE	R ² Score
9	0.3290	0.9928
10	0.4174	0.9908
11	0.5212	0.9885
12	1.5514	0.9669
13	21.6960	0.5868
14	2005.4662	-36.1771
15	13031132933.7197	-302784136.3581
16	659762111431.7620	-11341531518.9879
17	68865998620.7922	-1331147134.9910
18	201876030353.2496	-4105329683.8666
19	192136592239.8859	-3735532737.8618
20	145320422894.3460	-2729046150.7092



Model Selection and Rationale

Based on the table and the graph, I observed the following:

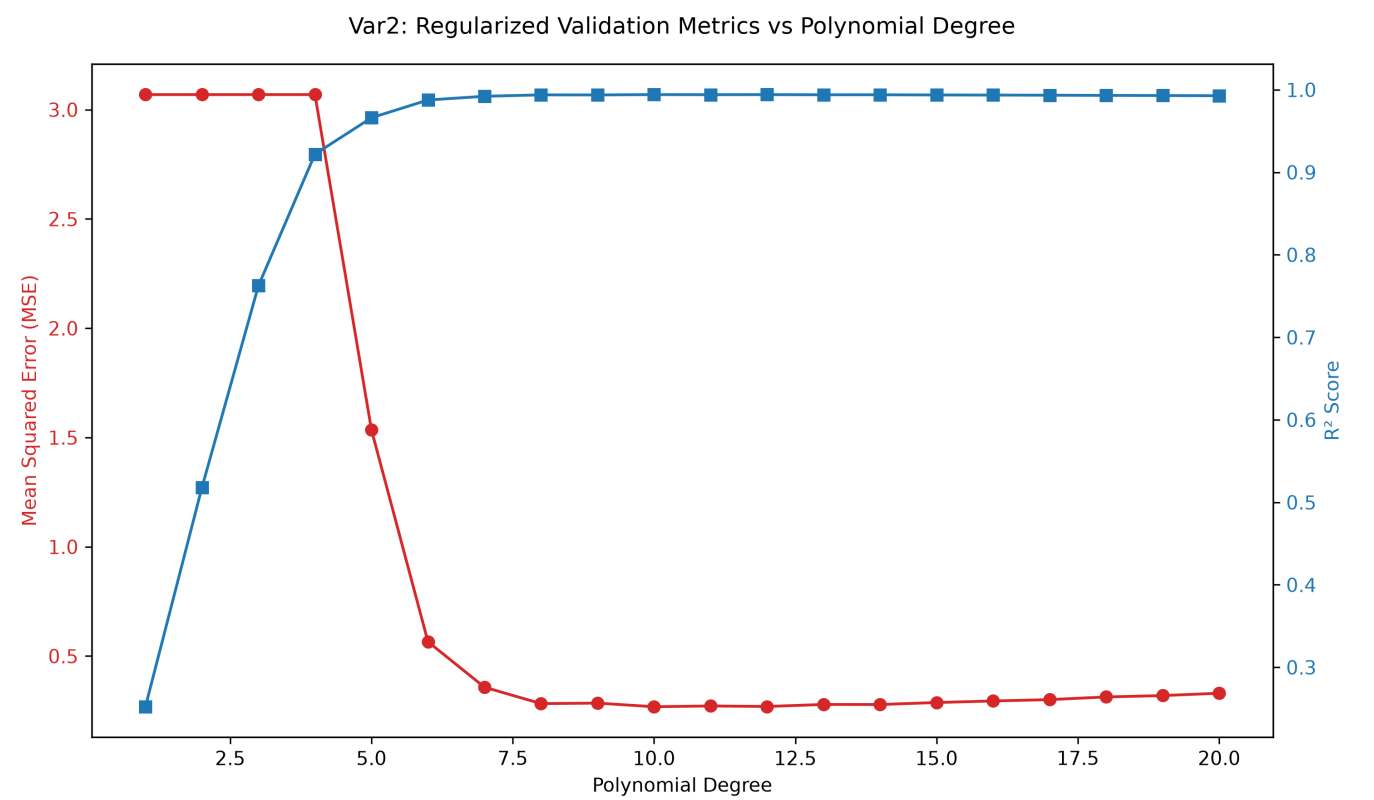
- 1. **Steady Improvement:** Unlike Phase 1, the Phase 2 data exhibited significantly higher complexity. As the polynomial degree increased from 1 to 8, the MSE continuously dropped and the R² score steadily improved.

- 2. **Optimal Fit at Degree 8:** My unregularized OLS model achieved peak validation performance at **Degree 8** (MSE = 0.2854, $R^2 = 0.9938$).
- 3. **Catastrophic Overfitting:** Starting from degree 9, performance slowly declined until degree 13 ($R^2 = 0.5868$). At degree 14 and beyond, I observed catastrophic overfitting, with the MSE exploding into the billions and R^2 scores dropping to massive negative numbers. The curse of dimensionality heavily affected these extremely high-degree polynomial models.

Regularization Testing (Ridge)

To stabilize the higher-degree models and extract better performance, I implemented Ridge Regression with Standard Scaling.

- **OLS Best:** Degree 8 (Val $R^2 = 0.9938$)
- **Ridge Best:** With Ridge regularization ($\alpha \approx 1.6238$), the model was able to successfully leverage a higher polynomial of **Degree 10**, pushing the peak validation R^2 to **0.9941**. Regularization also completely eliminated the catastrophic overfitting seen in degrees 14-20.



Conclusion for Phase 2: I decided the optimal model is a **Ridge Regularized Polynomial model of Degree 10 using all 3 features**, achieving the best overall performance and robustness.

Deliverables Generated

- **Prediction Files:**
 - `BT2024176_pred_var1.csv` (Updated with Degree 5 Ridge predictions)
 - `BT2024176_pred_var2.csv` (Updated with Degree 10 Ridge predictions)
- **Codebase:** `main.py` contains the final code using Ridge Regression and Standard Scaler to perform inference. `test_regularization.py` contains the advanced tuning logic.